Historical Partial Result: A Schatten Upper Bound for Raja's Covering Index

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Historical Status

This is the **superseded partial packet** that preceded the full Schatten sharp-decay packet. It proved the upper bound

$$\Theta_{S_p(H)}(n) \le n^{-1/p}$$

for infinite-dimensional Schatten classes. The parent packet adds the diagonal ℓ_p lower bound and therefore proves $\Theta_{S_p(H)}(n) \asymp n^{-1/p}$ for $1 < p < \infty$.

Source Gap

Kania–Maślany, arXiv:2512.13249, record lower bounds for non-commutative L_p-spaces and say that corresponding upper bounds are not pursued:

> **Proposition 5.2.** *Let (M, τ) be a semifinite von Neumann algebra such that $L_p(M, \tau)$ is infinite-dimensional, let $1 < p < \infty$, and set $r = \min\{p, 2\}$. Then there exists $c_p(M) > 0$ such that*
>
> $$\Theta_{L_p(M,\tau)}(n) \;\geqslant\; c_p(M)\, n^{-1/r} \qquad (n \geqslant 1).$$
>
> Thus the covering index of any non-commutative L_p-space decays at least like $n^{-1/r}$ with $r = \min\{p, 2\}$.
>
> *Remark* 5.3. If $M = L_\infty(\Omega, \mu)$ and $\tau(f) = \int_\Omega f\, d\mu$, then $L_p(M, \tau)$ is just $L_p(\mu)$ and Proposition 5.2 yields a lower bound of order $n^{-1/r}$ with $r = \min\{p, 2\}$. Our explicit covering in the commutative case (Proposition 3.1) shows that the optimal exponent is at most $1/p$; when $L_p(\mu)$ moreover admits an equivalent p-AUS norm, Corollary 3.3 gives $\Theta_{L_p(\mu)}(n) \asymp n^{-1/p}$. Thus the abstract non-commutative argument is not sharp even in the commutative setting when $p > 2$.
>
> If $M = B(H)$ with the canonical trace, then $L_p(M, \tau)$ is the Schatten class S_p on H. Proposition 5.2 then shows that
>
> $$\Theta_{S_p}(n) \;\gtrsim\; n^{-1/r}, \qquad r = \min\{p, 2\}.$$
>
> We do not pursue corresponding upper bounds in the non-commutative setting here.

They also state that sharp covering-index decay remains open even for the Schatten classes:

$$\rho_{L_p(M,\tau)}(t) \le C_p t^r, \qquad r = \min\{p, 2\},$$

and hence $L_p(M, \tau)$ is asymptotically uniformly smooth of power type $r > 1$. In particular, the natural norm on $L_p(M, \tau)$ is r-AUS with $r = \min\{p, 2\}$. This is the input used in the proof of Proposition 5.2: once $L_p(M, \tau)$ is r-AUS, Raja's general theorem (Theorem 1.6) yields a power-type lower bound

$$\Theta_{L_p(M,\tau)}(n) \gtrsim n^{-1/r}.$$

Determining the *precise* optimal AUS exponent and the corresponding sharp covering-index decay rate for specific non-commutative L_p-spaces remains an interesting open problem, even for classical examples such as the Schatten classes $S_p = L_p(B(H), \mathrm{Tr})$.

Idea of the Partial Proof

The scalar L_p proof in the source paper covers the unit ball by checking which piece of a measurable partition has small L_p-mass. For Schatten classes, the analogous partition is an orthogonal decomposition

$$H = H_1 \oplus \cdots \oplus H_n$$

and the coordinate projection is the corner compression $T \mapsto P_k T P_k$. Pinching onto diagonal corners is contractive on S_p, so one corner must have norm at most $n^{-1/p}$. The finite-codimensional inradius bound is forced by choosing a rank-one operator far out in that same corner, annihilating the finitely many functionals defining the finite-codimensional subspace.

Historical Theorem

Theorem 1. *Let H be an infinite-dimensional Hilbert space and let $1 \le p < \infty$. Then for every $n \in \mathbb{N}$,*

$$\Theta_{S_p(H)}(n) \le n^{-1/p}.$$

Moreover, the covering below has all pieces of essential inradius exactly $n^{-1/p}$.

Proof. Fix $n \in \mathbb{N}$, and write

$$H = H_1 \oplus \cdots \oplus H_n$$

as an orthogonal direct sum of infinite-dimensional closed subspaces. Let P_k be the orthogonal projection onto H_k, and define

$$C_k(T) = P_k T P_k, \qquad T \in S_p(H).$$

Put $a = n^{-1/p}$, and define

$$A_k = \{T \in B_{S_p(H)} : \|C_k(T)\|_p \le a\} \qquad (1 \le k \le n).$$

Each A_k is closed, convex, and bounded.

Let $E(T) = \sum_{k=1}^n P_k T P_k$ be the finite pinching. If $U_\varepsilon = \sum_{k=1}^n \varepsilon_k P_k$, where $\varepsilon \in \{-1, 1\}^n$, then

$$E(T) = 2^{-n} \sum_{\varepsilon \in \{-1,1\}^n} U_\varepsilon T U_\varepsilon.$$

The Schatten norm is unitarily invariant and convex, so $\|E(T)\|_p \le \|T\|_p$. Since $E(T)$ is block diagonal,

$$\|E(T)\|_p^p = \sum_{k=1}^{n} \|C_k(T)\|_p^p.$$

Thus every $T \in B_{S_p(H)}$ belongs to some A_k, and the A_k's cover the unit ball.

The lower bound $\varrho(A_k) \ge a$ is immediate from $aB_{S_p(H)} \subseteq A_k$. To prove $\varrho(A_k) \le a$, fix $T \in A_k$, a finite-codimensional subspace $Y \le S_p(H)$, and $r > a$. Put $V = C_k(T)$. Choose functionals $\Phi_1, \ldots, \Phi_m \in S_p(H)^*$ with

$$Y = \bigcap_{j=1}^{m} \ker \Phi_j.$$

By trace duality, and using $S_1(H)^* = B(H)$ at the endpoint, there are operators $B_j \in B(H)$ such that for rank-one operators $\theta_{u,v}(\xi) = \langle \xi, v\rangle u$ one has

$$\Phi_j(\theta_{u,v}) = \langle B_j u, v\rangle.$$

Choose a finite-rank projection $R \le P_k$ so that

$$\left(\|RVR\|_p^p + r^p\right)^{1/p} - \|V - RVR\|_p > a.$$

This is possible because finite-rank compressions approximate V in S_p-norm and the left side tends to $(\|V\|_p^p + r^p)^{1/p} > r > a$.

Let $K = H_k \ominus R(H_k)$. Pick a unit vector $u \in K$, then choose a unit vector

$$v \in K \cap \operatorname{span}\{B_1 u, \ldots, B_m u\}^{\perp}.$$

Set $W = \theta_{u,v}$. Then W is supported in the k-th corner, $\|W\|_p = 1$, and $W \in Y$. Moreover RVR and W are block diagonal with respect to orthogonal summands, so

$$\|RVR + rW\|_p^p = \|RVR\|_p^p + r^p.$$

Therefore

$$\|C_k(T + rW)\|_p = \|V + rW\|_p \ge \|RVR + rW\|_p - \|V - RVR\|_p > a.$$

Hence $T + rW \notin A_k$. No finite-codimensional ball of radius larger than a can sit inside A_k, so $\varrho(A_k) \le a$. □

Supersession

This upper-bound result was upgraded in the parent packet by observing that the diagonal copy of ℓ_p is contractively complemented in $S_p(H)$. Together with monotonicity of Θ for contractively complemented subspaces and the scalar estimate $\Theta_{\ell_p}(n) \asymp n^{-1/p}$, this gives the full sharp Schatten decay for $1 < p < \infty$.

References

[1] T. Kania and N. Maślany, *Raja's covering index of L_p spaces*, arXiv:2512.13249.